

MEHMET EMİN BALOĞLU

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SUMMARY

AI Engineer & Data Scientist with 3+ years of experience in Computer Vision and LLM-based systems for industrial safety. Proven in deploying end-to-end ML pipelines and optimizing production models, reducing false positives by up to 22% across 70+ industrial zones. Currently focused on agentic workflows and modular RAG systems for next-generation AI applications.

KEY AREAS OF EXPERTISE

- **Core Domains:** Computer Vision, NLP, Generative AI, LLMs, GANs, AI & Agentic Workflows
- **Specializations:** Object & Anomaly Detection, Image Processing, Transfer Learning, Model Optimization
- **Programming & Data:** Python, Pandas, NumPy, SQL
- **Frameworks:** LangChain, LangGraph, LlamaIndex, Smolagents, RAG Architectures
- **Metrics & MLOps:** Model Evaluation (mAP, F1-score), Automation Pipelines
- **Generative AI:** Flux, Stable Diffusion, Gemma, Ollama
- **Databases & Retrieval:** Vector Databases (ChromaDB), Graph Databases (Neo4j)
- **LLMs & NLP:** Large Language Models (LLMs), NLP, RAG, Graph-RAG, Hybrid Retrieval
- **Tools:** Docker, Git, Linux, VS Code, Jupyter

PROFESSIONAL EXPERIENCE

Event Gates | Bursa, Turkey

Event Gates is a Bursa-based industrial AI company specializing in machine vision technologies for occupational health and safety. Through its core products, Safety AI and Quality AI, the company delivers real-time risk detection and quality control systems integrated with complex industrial machines. Serving over 70 industrial zones across Turkey, including clients such as Renault, Tofaş, and Assan, Event Gates aims to become a global leader in AI-powered industrial safety and inspection solutions.

AI Engineer & Team Lead | 05/2022 – 10/2025

Computer Vision • Data Science • AI Research

- Spearheaded the design and development of tailored deep learning and computer vision solutions, addressing real-world challenges in occupational health and safety.
- Worked with large-scale video streams, industrial ip camera feeds, and sensor-triggered event data in safety-critical environments.
- Led full-cycle AI workflows — from data collection, cleaning, and annotation to model training, evaluation, and optimization — ensuring high-performance and reliable outcomes.
- Engineered robust Python scripts to support model deployment, inference pipelines, and task automation, streamlining operational efficiency across projects.
- Optimized YOLOv4-based production models using Optuna-driven hyperparameter tuning, reducing false positives in industrial safety systems.
- Collaborated closely with field engineers, product teams, and safety experts to iterate on model requirements and validation criteria.
- Achieved up to 22.6% FP reduction on a 670K+ image custom dataset while maintaining mAP and F1-score, and integrated results into standard training pipelines.
- Conducted R&D on cutting-edge AI technologies, creating proof-of-concept demos and technical documentation to guide internal stakeholders and future development.

EDUCATION

Firat University | Elazığ, Turkey

Bachelor's Degree, Mechatronics Engineering | GPA: 3.07/4.0 | 2017 – 2020

CERTIFICATIONS & COURSES

IBM AI Engineering Professional Certificate (Umbrella for the IBM courses below) 

- Machine Learning with Python (IBM) 
- Deep Learning & Neural Networks with Keras (IBM) 
- Building Deep Learning Models with TensorFlow (IBM) 
- Deep Neural Networks with PyTorch (IBM) 
- Introduction to Computer Vision and Image Processing (IBM) 
- **Huawei Cloud Practicum** 
- **LLM Mastery from Udemy** 
- **Agents Course from HuggingFace** 
- **Diffusion Mastery: Flux, Stable Diffusion Course from Udemy** 

PROJECTS:

- **Estate-Agent – RAG & Hybrid Retrieval-Based Conversational AI Agent**
Developed a conversational real-estate agent leveraging Retrieval-Augmented Generation (RAG) to interpret natural language queries and provide relevant property recommendations. Integrated vector retrieval using ChromaDB within a hybrid search architecture and structured metadata filtering to enable precise, location-aware responses for the Bursa real estate market. Built with Python, LLaMA-based models, and a Gradio interface. (Github: )
- **Modular LLM-Powered Multitask AI Agent**
Built a modular multimodal AI agent using Gemma 3 and SmolAgent for the Hugging Face AI Agents course, designed to answer prompts from the GAIA benchmark, enabling instruction-following and reasoning across documents, images, audio, and YouTube videos with transcript extraction. (Github: )
- **AI-Powered Background Generator**
Developed a web app that removes and regenerates image backgrounds using Stable Diffusion models. Designed with Gradio for user interaction.
(Github: )
- **Animal Image Classifier with Gradio**
Designed and deployed an end-to-end image classification system including preprocessing, feature extraction, clustering, transfer learning, and a Gradio-based frontend.
(Github: )
- **Bus Capacity Prediction Platform**
Created a machine learning system that predicts bus occupancy rates using a MongoDB dataset. Built and deployed the system with a Django web interface.
(Github: )

LANGUAGES

- **Turkish:** Native
- **English:** Intermediate (Professional Reading & Technical Communication)